

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

0862/01

Paper 1

October 2025

1 hour

You must answer on the question paper.

You will need: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages.

- 1 Work out.

$$35.8 \times 10^3$$

..... [1]

- 2 Draw a ring around the **two** calculations that will give a positive answer.

$$-5 \times 3$$

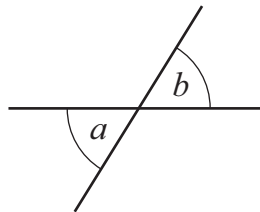
$$(-4)^2$$

$$-4 \times -2$$

$$16 \div -2$$

[1]

- 3 The diagram shows two intersecting lines.



Angelique says, ‘Angle a and angle b are corresponding angles.’
Angelique is **not** correct.

Complete the sentence with the correct description of the angles.

Angle a and angle b are angles.

[1]

- 4 Ahmed wants to find out the amount of money students of different ages spend on lunch and which food is the most popular.

Give **one** example of quantitative data he should collect.

..... [1]

- 5 Write the correct numbers in the boxes to complete the calculation.

$$0.84 \div 0.12 = \boxed{} \div 12 = \boxed{}$$

[2]

- 6 Use compasses and a ruler to construct a triangle with all three interior angles of 60° .
The base has been drawn for you.
Do **not** rub out your construction arcs.



[2]

- 7 The cost of hiring a car is a charge of \$50 per day plus a fixed cost of \$80
A car is hired for n days.

Draw a ring around the formula that represents the total cost, \$ C , of hiring this car.

$$C = 50 + 80n$$

$$C = 50n \times 80$$

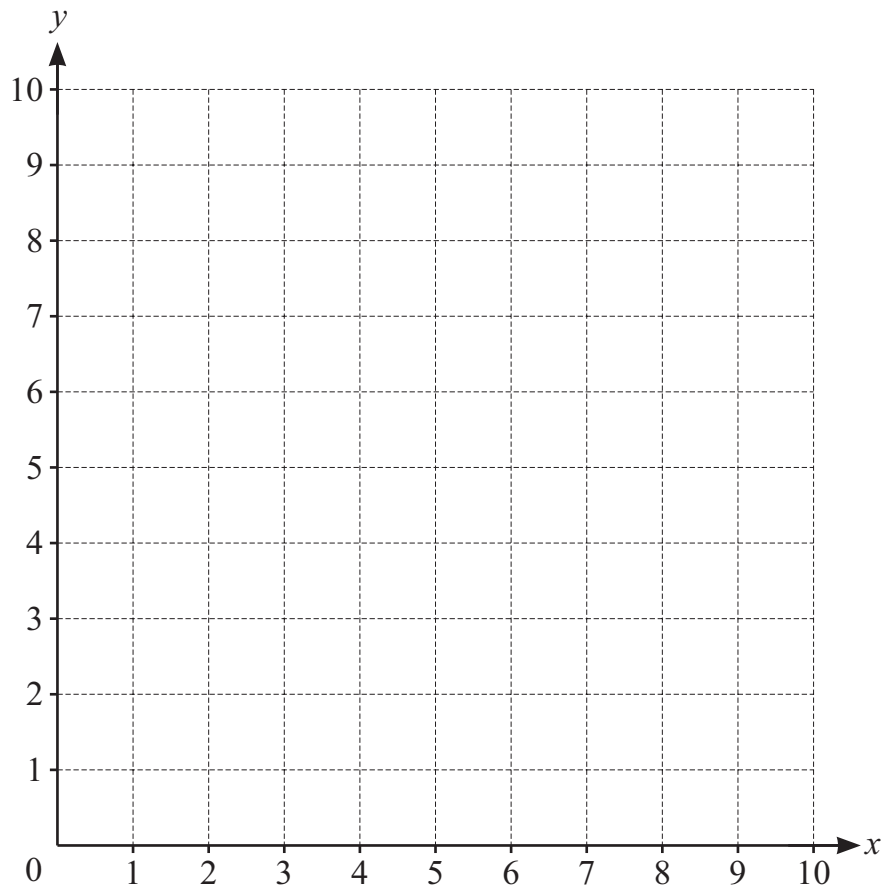
$$C = (50 + 80)n$$

$$C = 50n + 80$$

[1]

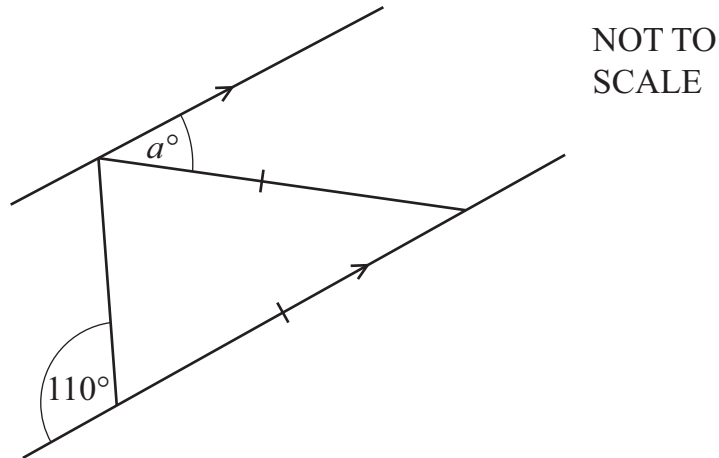
- 8 Draw the graph of $4x + 5y = 40$
You may use the table to help you.

x	0	
y		0



[3]

- 9 The diagram shows an isosceles triangle with its vertices on two parallel lines.



Find the value of a .

$a =$ [3]

- 10 Draw a ring around the correct calculation to **increase** 88 by 13%.

88×0.13

13×0.88

13×1.88

88×1.13

$88 \div 0.13$

[1]

- 11 A prism has a cross-section that is a **regular** polygon.
The prism has 4 planes of symmetry.

Write down the mathematical name for the shape that is the cross-section of the prism.

..... [1]

12 (a) Simplify $x^4 \times x^5$

..... [1]

(b) Write an integer in each box to complete the statement.

$$\boxed{} y^{15} \div 3y \boxed{} = 6y^8$$

[2]

13 Use one of the symbols $>$, $<$ or $=$ to complete each of these statements.

1 000 000 micrometres 1 metre

100 nanometres 1 micrometre

1 000 000 kilometres 1 light year

[2]

14 Eva walks the length of her garden.

She takes 20 seconds walking at a speed of 4 km/h.

Find the number of seconds she takes to walk the length of her garden at a speed of 5 km/h.

..... seconds [2]

15 Find the value of $5(x + 2)^2 - 7$ when $x = -3$

..... [2]

16 The table shows some information about the heights of children in class A and class B.

	Class A	Class B
Mean	1.40 m	1.42 m
Range	0.15 m	0.12 m

Tick (✓) to show if each of these statements is true or false or you cannot tell.

	True	False	Cannot tell
Children in class A are generally taller because the range for class A is larger.	☐	☐	☐
Children in class B are generally taller because the mean for class B is larger.	☐	☐	☐
The tallest child is in class B.	☐	☐	☐

[2]

17 Solve.

$$\frac{40}{x} = 20$$

$$x = \dots\dots\dots$$

$$\frac{8}{y-3} = 2$$

$$y = \dots\dots\dots$$

[3]

18 For this question, x can be an integer or a decimal.

Tick (✓) to show if each of these statements is true or false.

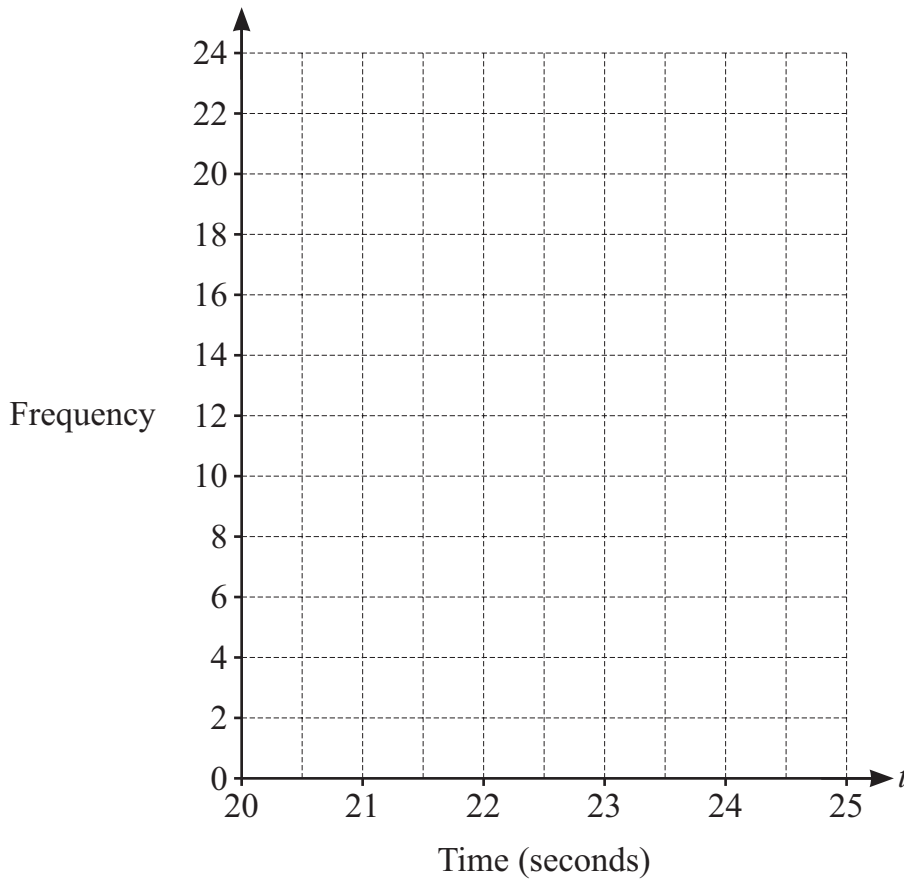
Statement	True	False
$x < -2$ is equivalent to $-2 < x$		
$5 > x$ is equivalent to $12x < 60$		
$x \geq 3$ is equivalent to $x > 2$		
$4x \geq 12$ is equivalent to $x > 3$		

[2]

19 The table shows information about the time taken for some athletes to complete a race.

Time (t seconds)	Frequency
$20 < t \leq 21$	5
$21 < t \leq 22$	12
$22 < t \leq 23$	17
$23 < t \leq 24$	22
$24 < t \leq 25$	2

(a) Draw a frequency polygon to show this information.



[3]

(b) The slowest time taken to complete the race was 24.1 seconds.

Draw a ring around each of the possible values for the range of the times taken in seconds.

5

4.2

4

3.1

3

[2]

20 A sequence has a first term of 22

The term-to-term rule is to subtract $2\frac{1}{2}$

(a) Show that the 10th term in this sequence is negative.

[2]

(b) Draw a ring around each value that is a term in this sequence.

-8

-12

-23

-47

-88

[2]

21 Here is a number fact.

$$102\,292 \div 428 = 239$$

Use this number fact to complete these calculations.

$$4.28 \times 23.9 = \dots\dots\dots$$

$$1022.92 \div 0.0239 = \dots\dots\dots$$

[2]

22 Work out.

$$2\frac{7}{25} \times \left(9.5 + \frac{1}{2}\right)^3$$

..... [3]

- 23** Point A has the coordinates $(-9, -4)$ and point C has the coordinates $(21, 11)$.
Point B lies on the line AC .
 $AB : BC = 2 : 3$

Find the coordinates of point B .

(..... ,) [3]

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